

main.py

```
1 from collections import deque
2
3 def bfs(graph, start):
4     visited = set()
5     queue = deque([start])
6     visited.add(start)
7     order = []
8
9     while queue:
10         vertex = queue.popleft()
11         order.append(vertex)
12
13         for neighbor in graph[vertex]:
14             if neighbor not in visited:
15                 visited.add(neighbor)
16                 queue.append(neighbor)
17
18     return order
19
20
21 def dfs(graph, start, visited=None, order=None):
22     if visited is None:
```

```
22     if visited is None:
23         visited = set()
24
25     if order is None:
26         order = []
27
28     visited.add(start)
29     order.append(start)
30
31     for neighbor in graph[start]:
32         if neighbor not in visited:
33             dfs(graph, neighbor, visited, order)
34
35     return order
36
37
38 # Representing graph as an adjacency list
39 graph = {
40     'A': ['B', 'C'],
41     'B': ['A', 'D', 'E'],
42     'C': ['A', 'F'],
43     'D': ['B'],
```

```

30
31     for neighbor in graph[start]:
32         if neighbor not in visited:
33             dfs(graph, neighbor, visited, order)
34
35     return order
36
37
38 # Representing graph as an adjacency list
39 graph = {
40     'A': ['B', 'C'],
41     'B': ['A', 'D', 'E'],
42     'C': ['A', 'F'],
43     'D': ['B'],
44     'E': ['B', 'F'],
45     'F': ['C', 'E']
46 }
47
48 print("BFS Traversal starting from 'A':", bfs(graph, 'A'))
49 print("DFS Traversal starting from 'A':", dfs(graph, 'A'))
50 print("Name: Moulya M P")
51 print("Register Number: 24BECS167")

```

| Run | Output                                                                                                                                                                                                                                |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     | <pre> ^ BFS Traversal starting from 'A': ['A', 'B', 'C', 'D', 'E', 'F']   DFS Traversal starting from 'A': ['A', 'B', 'D', 'E', 'F', 'C']   Name: Moulya M P   Register Number: 24BECS167    === Code Execution Successful === </pre> |